

Chapter 3

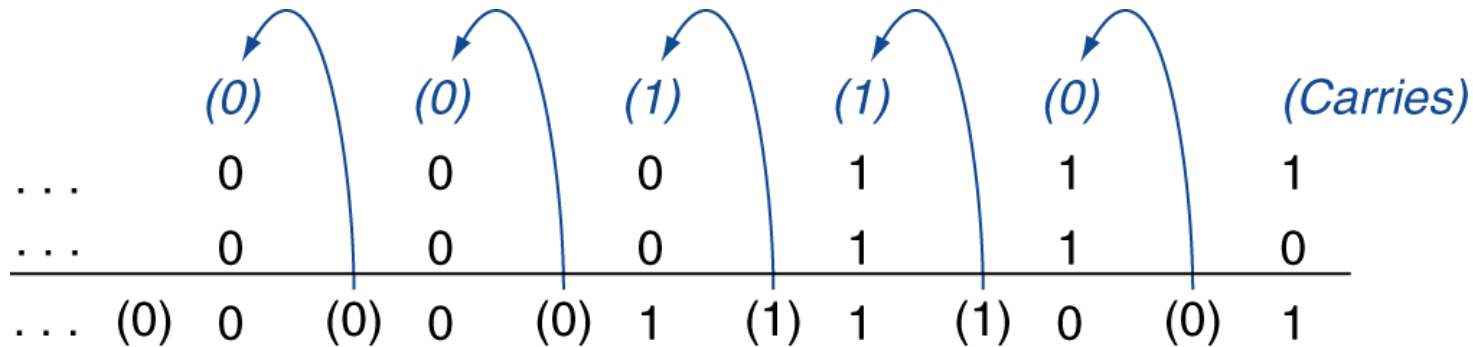
Arithmetic for Computers

3.1 Arithmetic for Computers

- Operations on integers
 - Addition and subtraction
 - Multiplication and division
 - Dealing with overflow
- Floating-point real numbers
 - Representation and operations

3.2 Integer Addition

■ Example: $7 + 6$



■ Overflow if result out of range

- Adding +ve and -ve operands, no overflow
- Adding two +ve operands
 - Overflow if result sign is 1
- Adding two -ve operands
 - Overflow if result sign is 0

3.2 Integer Subtraction

- Add negation of second operand
- Example: $7 - 6 = 7 + (-6)$

+7:	0000 0000 ... 0000 0111
-6:	1111 1111 ... 1111 1010
+1:	0000 0000 ... 0000 0001

- Overflow if result out of range
 - Subtracting two +ve or two -ve operands, no overflow
 - Subtracting +ve from -ve operand
 - Overflow if result sign is 0
 - Subtracting -ve from +ve operand
 - Overflow if result sign is 1